

5. Analytical Modelling of Parallel Programs *notes*

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1 Performance Metrics

Total parallel overhead

$$T_o = pT_p - T_s$$

Speedup

$$S = \frac{T_s}{T_p}$$

$$S \leq p$$

Efficiency

$$E = \frac{S}{p}$$

$$1 \leq E \leq p$$

some terms: cost optimal = pt_p -optimal $\implies E \sim \Theta(1)$
we define problem size as the number of basic computation steps in the best sequential algorithm to solve the problem on a single processing element, where the best sequential algorithm is defined as in Section 5.2.3. Because it is defined in terms of sequential time complexity, the problem size is a function of the size of the input. The symbol we use to denote problem size is W .

$$T_p = \frac{W + T_o(W, p)}{p}$$
$$S = \frac{pW}{W + T_o(W, p)}$$
$$E = \frac{1}{1 + T_o(W, p)/W}$$

We need to maintain the ratio T_o/W for keeping given efficiency a constant E as given by the following equation

$$\frac{T_o(W, p)}{W} = \frac{1 - E}{E}$$

$$W \propto T_o \quad \text{for constant } E$$

$$W = KT_o(W, p) \quad \text{as } K = E/(1 - E)$$

$$W = W(p) \quad \text{for keeping } E \text{ constant, this is called as } \mathbf{isoefficiency \text{ function}}$$

cost optimality and isoefficiency function

$$pT_p = \Theta(W)$$

$$W + T_o(W, p) = \Theta(W)$$

$$T_o(W, p) = O(W)$$

$$W = \Omega(T_o(W, p))$$

when we take into account the maximum degree of concurrency of system $C(W)$, the equation for parallel time becomes

$$T_p = \frac{W + T_o(W, C(W))}{C(W)}$$